AI4LA: An Intelligent Chatbot for Supporting Students with Dyslexia, Based on Generative AI

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Abstract. In the last two years, thanks to the advent of generative AI, new avenues are opening up in its use in the educational field, e.g. through the use of ChatGPT. But, while generative AI is opening up new avenues for both teaching and learning, there are no proposals for its use in the context of students with Dyslexia and in general for students with Specific Learning Disorders (SLD), that afflict between 10 and 15 percent of people worldwide. In this paper we present AI4LA, a web intelligent chatbot based on generative AI, with the aim to help dyslexic students during their learning process. This system is capable of analyzing conversational data to model students' understandings and misconceptions, thereby offering personalized educational support. Significantly, we highlight the chatbot's ability to create Concept Maps from conversations, visually representing the student's knowledge structure and progression. An evaluation of the effectiveness of the system is presented, in a real-world educational settings, based on feedback from a sample of dyslexic students. The findings suggest that our system holds great promise in revolutionizing the way dyslexic students engage with educational content, making learning more accessible, interactive, and effective.

Keywords: Dyslexia · Intelligent Chatbot · LLM

1 Introduction

The shift towards distance learning has gained momentum in recent years, especially due to the COVID-19 pandemic, leading to an increase in the adoption of Massive Open Online Courses (MOOCs) across all educational levels [7]. This transition has posed significant challenges in providing personalized support to students, particularly those with Specific Learning Disorders (SLDs) like *Dyslexia*, *Dysorthographia*, and *Dyscalculia*, who face substantial hurdles in environments lacking direct teacher engagement [12,14]. Dyslexia, the most prevalent SLD, affects between 5% and 15% of the global population, impacting reading, writing, and occasionally spoken language comprehension [1,2]. The development

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A. Sifaleras and F. Lin (Eds.): ITS 2024, LNCS 14798, pp. 369–377, 2024.
https://doi.org/10.1007/978-3-031-63028-6_31

of assistive technologies such as speech recognition, text-to-speech, and smart pens has been crucial in supporting dyslexic learners [3,6,11,16]. Furthermore, advances in Deep Learning and Large Language Models (LLMs) like ChatGPT have facilitated the creation of tools like the system we present here: *AI4LA-Artificial Intelligence for Learning Assistant*, which supports dyslexic students by adapting dialogues and generating personalized Concept Maps (CM)s based on their specific needs[1]. Our initial trials of *AI4LA* in a high school setting have shown promising results, enhancing engagement and learning outcomes for students with dyslexia. This paper details these findings, discusses the integration of generative AI in educational settings, and outlines future research directions

2 Related Work

This section examines support systems for individuals with dyslexia, particularly highlighting the *AI4LA* system's use of CMs, which visually represent a knowledge domain as a direct graph, formed by nodes (the concepts) and the connections among them (the semantic relationships). These CMs, dynamically generated from chatbot-learner dialogues, help minimize the reliance on textual explanations, addressing the challenges dyslexic learners face in creating such maps due to difficulties with reading and key relationship identification [8].

Beyond CMs, *AI4LA* leverages advanced generative AI technologies like *Retrieval Augmented Generation* (RAG) to enhance learning by generating topic summaries, distinguishing it from other tools that lack dynamic CM customization capabilities. Comparative analysis shows that other applications also support dyslexic learners. The ALEXZA application [13] offers features like text-to-speech and adjustable reading speeds, using machine learning to classify text difficulty. The GARY system enhances reading through eye-tracking and auditory narration, significantly improving comprehension [15]. Additionally, the RALF framework uses Cellular Learning Automata (CLA) to tailor educational content for Persian students with dyslexia, enhancing learning effectiveness by 27% during COVID-19 lockdowns [10]. In [5], the TutorChat system is presented. This system is a chatbot based on ChatGPT, able to automatically build multimedia CMs: each node is connected to images and video concerning the knowledge domain under study. Our system, however, offers different types of CMs whereas Tutorchat only builds them based on the answer to a specific question using Google for images and youtube for videos.

Overall, the *AI4LA* system's integration of generative AI and CMs uniquely positions it in the educational support landscape for dyslexic learners, setting new standards for personalized educational tools.

3 The System

In the rapidly evolving landscape of Natural Language Processing (NLP), two powerful techniques have emerged as game-changers, reshaping the way we harness the potential of language models: *Fine-Tuning* [4] and *Retrieval Augmented*

[1] https://dyslexia2024.replit.app.

Generation (RAG) [9]. These methodologies have revolutionized the capabilities of language models, addressing critical limitations and unlocking new possibilities in various NLP applications. Since our system uses both of the above technologies, it is worth briefly summarising their key features and how were used for a full understanding of the system.

Fine Tuning is essentially a re-training of a Deep Learning model. In the context of refining LLMs, Adaptive Fine-Tuning (AFT) and Behavioral Fine-Tuning (BFT) refer to specific methodologies used to adjust and improve the model based on additional data or feedback. AFT involves further training a pre-trained model on a specific, often smaller, dataset that is tailored to the needs of a particular application or domain, adapting the model to be more proficient in areas that might not have been adequately covered during the initial broad training phase. BFT, on the other hand, focuses on adjusting the model's outputs to match desired behaviors more closely, often in response to specific user feedback or desired interaction styles. This method is less about domain knowledge and more about how the model interacts-its style, tone, response formulation and so on. In our case, the fine-tuning process we employed is an example of BFT. The dataset we used was specialized on a set of simulated conversations between the chatbot and dyslexic students, with the aim to train the chatbot to interact properly with students with dyslexia. This work was supported by a cognitive psychologist. The dataset is composed of 80 conversations, in the form: $C_i \equiv \{s_i, (u_{ij}, a_{ij})\}$, where s_i (i=1,...,80) is that part of the conversation used to train the role of the system and (u_{ij}, a_{ij}) are a set of interactions in the form: $<$*question posted by the user*,*answer prompted by the chatbot*$>$. So, while each record C_i has only one s_i, it can have a variable number of couples (u_{ij}, a_{ij}), depending on the knowledge domain concerning the interaction. Figure 1 shows an example of such kind of record with $j = 2$.

system	Your role is to be a helpful assistant with a friendly, understanding, patient, and user-affirming tone. You should: explain topics in short, simple sentences; keep explanations to 2 or 3 sentences at most. If the user provides affirmative or brief responses, take the initiative to continue with relevant information. Check for user understanding after each brief explanation using varied and friendly-toned questions. Use ordered or unordered lists (if longer than 2 items, introduce them one by one and check for understanding before proceeding), or simple text in replies. Provide examples or metaphors if the user doesn't understand.
user	Tell me something about Planet Earth
assistant	Planet Earth is the third planet from the Sun in our Solar System. It's the only known planet to support life. Are you following?
user	Yes, please continue.
assistant	It has diverse climates and terrains, from icy polar regions to vast deserts. About 71% of its surface is covered by water, mainly oceans. Clear so far?
user	Absolutely clear.

Fig. 1. A complete record of the fine-tuning dataset. In this example, first we train the chatbot to have the role of an assistant to answer to dyslexic people. Second, we have two simulated interactions chatbot-user. In this case, $j = 2$.

In this way, we obtained a neural model customized to properly interact with dyslexic people.

RAG provides a way to optimise the production of an LLM with targeted information without changing the model behind it; that targeted information can be more up-to-date than the LLM, but also more precise in the case of specific organisations and sectors. This means that the generative AI can provide more appropriate responses to prompts, and base those responses on highly current data [9]. A RAG model acts in a pipeline configuration, following these steps: Retrieve, Augment and Generate.

The system RAG pipeline uses *Chroma DB*[2]. Chroma DB is an open-source vector store designed for storing and retrieving vector embeddings. Its primary function is to save embeddings along with metadata for use later by large LLMs. It supports the creation, storage, and querying of embeddings, making it a valuable tool for developers building AI applications that require efficient handling of vector embeddings, as in the case of our system. Figure 2 shows the functional architecture of the system.

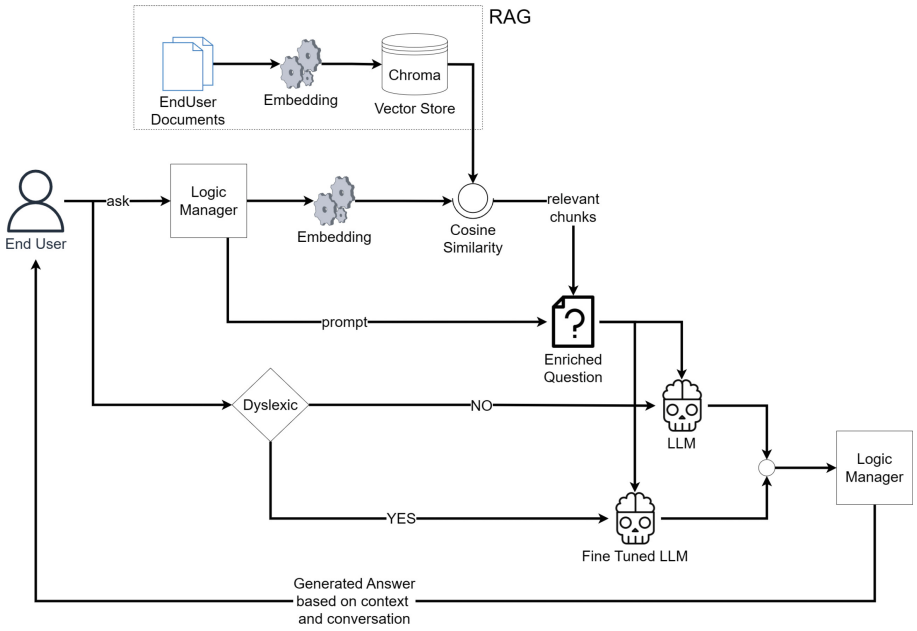


Fig. 2. The Fine-Tuned RAG Pipeline embedded in the AI4LA system. For dyslexic people the system makes use of the fine-tuned model.

In particular, our system empowers users to gather data from various sources such as PDFs, text documents, and YouTube videos. Subsequently, this collected

[2] https://www.trychroma.com/.

data are utilized to generate a Chroma DB collection, which serves as valuable context for enhancing the capabilities of the RAG model. When a user submits a query, the RAG system uses the indexed data and vectors to perform efficient searches. The system identifies relevant information by comparing the query vector with those stored in Chroma DB. In our implementation we used the cosine similarity measure to find the most relevant chunks inside the vector DB. The LLMs then use the retrieved data to craft appropriate responses. Finally, we have enriched the RAG Pipeline with those finely-tuned models specifically designed to meet the unique requirements of dyslexic users and discussed in the previous paragraph.

The system offers a range of features which ensure that every interaction is tailored to the student's unique preferences and needs: (i) a typeface designed by Google, against some common symptoms of dyslexia; (ii) Context Summarization: When the learner uploads her own context, the system enables her to request a summary. This summary provides an overview of the context's key points, ensuring the learner has a clear understanding of the topic at hand during conversations; (iii) Contextual Concept Maps: When the chatbot provides an answer, the user can generate a CM for that response; (iv) Speech-to-Text: The student can post the question using voice instead of writing and the system will transcribe it into text, streamlining communication; (v) Export of Full Conversation: The system allows users to conveniently export the entire conversation, preserving it for reference or future analysis. In addition, we placed a significant emphasis on user customization. In fact, users have the power to adjust the level of contextual CM generation and conversation memory. This allows the chatbot to fully adapt to their needs and preferences, making the interaction even more personal and tailored to the user.

Here we show a brief case study of the chatbot at work. After entering the system using his credentials, the (dyslexic) student entered a PDF document into the system, concerning operations between binary numbers. He then asked the following question: *Please show me an example of addition of two binary numbers*. Consequently, the chatbot prompts the example. After that, the student asks for the CM that is immediately drawn in the space below the answer, as shown in Fig. 3. Finally, the student can download the CM in order to better fix the concepts to be studied.

4 Methods

This section details the initial experimentation with the system, which involved a sequence of steps: selecting a sample, administering a text comprehension test, and distributing a happiness questionnaire. The study included 23 dyslexic students, aged 14 to 18, from a technical high school in Italy. The test chosen was based on a newspaper article titled *His Majesty the Owl Blinded by Lights*, selected due to its relevance to text comprehension-a critical skill for dyslexic students-and because it mimics the standardized *Invalsi Test* used in Italian education. Students used the intelligent chatbot pre-loaded with the text to

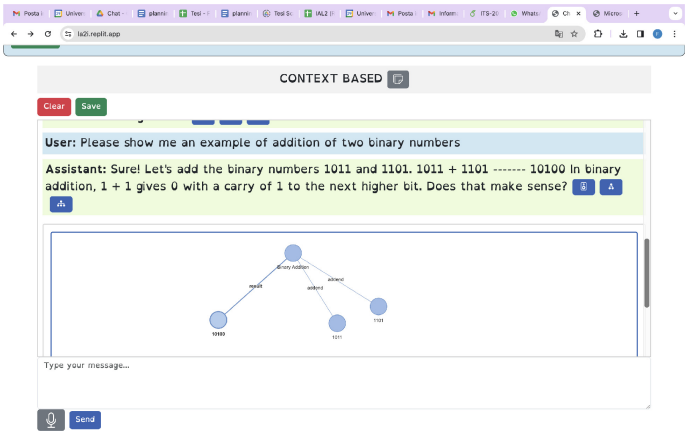


Fig. 3. The brief case study concerning the learning of binary numbers.

assist them in understanding and answering questions about the article. They had 45 min to complete the test in a fully anonymized setting to protect their privacy, using unique credentials and pseudonyms. Prior to the test, students were briefed on how to use the chatbot, despite their previous general experience with ChatGPT. This was their first opportunity to query about self-provided texts, a feature enabled by this system. Both paper and digital versions of the test were available, and each student worked at a dedicated PC station. After the test, they completed a feedback questionnaire. Data analysis was performed on both the chatbot interactions and questionnaire responses.

In the analysis of the data generated by the experiment, we distinguished two keys of study the first concerns an analysis of student-system interactions and utilization, as retrieved from the log file. The second concerns the questionnaire taken by the sample after the session.

Student-System Interactions. Our analysis identified four key variables in student interactions with the system: Question Precision, Seeking Clarifications, Use of CMs, and Low Usage. From the data (see Table 1), we observed that slightly more than 50% of students asked *Precise Questions*, indicating that the chatbot is a valuable resource for direct answers, aiding in test preparation. Approximately 20% sought clarifications or engaged more deeply with the chatbot's responses, demonstrating its role in fostering an interactive learning environment. Notably, 65% of students utilized the CMs generated by the system, showing the effectiveness of this feature in helping students visualize and organize information. However, a small segment, about 5%, showed *Low Usage* of the system, either by not using it or interacting minimally. Investigating the reasons behind this could provide insights into potential improvements in system design and functionality. Overall, the chatbot appears to effectively meet the needs of most students by providing direct answers, enabling deeper

understanding, and supplying helpful visual aids, though further research could enhance its appeal to all users.

Table 1. The four variables on which the system utilisation analysis was based.

Chat Activities	%
Question Precision	50%
Seeking Clarifications	20%
Use of CMs	65%
Low Usage	5%

The Questionnaire. After the work session, all students completed a questionnaire designed to assess their enjoyment, engagement, and the usability of the system. The questionnaire was distributed using the Google Form module[3] and consisted of sixteen closed-ended questions and one open-ended question. The questionnaire is depicted in Fig. 4. Crafted with the assistance of a psychologist specializing in dyslexia, the questions primarily focused on usability, engagement, and the potential effectiveness of *AI4LA* as a supportive tool for dyslexic learners.

ID	Question	ID	Question
Q_1	What is your age?	Q_9	Are the text characters with which AI4LA communicates clear and understandable to you?
Q_2	What kind of school you attend	Q_{10}	Do you think AI4LA can help you in your studies?
Q_3	Would you like to use IA4LA often to learn?	Q_{11}	Do you like that AI4LA can read your answer?
Q_4	Did you find IA4LA easy to use?	Q_{12}	Do you like that you can communicate with your voice without writing?
Q_5	Did you need technical help to use AI4LA?	Q_{13}	Did AI4LA's explanations help you?
Q_6	Did you feel confident using AI4LA?	Q_{14}	Did you find the concept map AI4LA proposed easy to read?
Q_7	Did you have to learn many technical things before you could start communicating with AI4LA?	Q_{15}	Did the concept map help you?
Q_8	Did you feel encouraged by AI4LA?	Q_16	Did you enjoy using AI4IA?

Fig. 4. The questions composing the questionnaire submitted to the sample after the use of the system.

The closed-ended questions were grouped according to three research themes:

– Group $G_1 = \{Q_4, Q_5, Q_6, Q_7, Q_8, Q_9, Q_{12}\}$ focused on assessing the system's ease of use.

[3] https://forms.gle/RNKCxDjuEyu1RjdU9.

- Group G_2=$\{Q_3, Q_8, Q_{10}, Q_{11}, Q_{12}, Q_{13}, Q_{14}, Q_{16}, Q_{15}\}$ aimed to evaluate the acceptance and appreciation of IA4LA as a study support tool.
- Group G_3=$\{Q_3, Q_6, Q_8, Q_9, Q_{10}, Q_{11}, Q_{14}\}$ intended to measure the perceived effectiveness of the system.

Responses were gathered using a 5-point Likert scale, with specific questions designed inversely to ensure analytical consistency. Normalization was applied to invert responses for Q_5 and Q_7 during analysis, aligning them with the standard scale direction used in other questions. Data analysis involved calculating the average number of occurrences for each scale value across the groups, facilitating a detailed evaluation of user feedback.

Table 2. Occurrences of the 5-scale values in the answers to the questionnaires, aggregated according to the three groups/RQs.

	Avg. 1	Avg. 2	Avg. 3	Avg. 4	Avg. 5
Group 1	0	0.78	1.8	4.23	8.98
Group 2	0	0.41	2.21	4.31	8.76
Group 3	0	0.22	1.9	4.78	8.89

Initial findings from this analysis indicate a lack of low-value responses, with the higher values on the scale dominating, suggesting a positive reception towards the *AI4LA* system. The detailed results from each group are summarized in Table 2, providing a clear indication of high user engagement and the perceived effectiveness of the system.

5 Conclusions

In this article, we presented *AI4LA*, an intelligent web chatbot based on generative AI to support the learning of dyslexic students. The system has the characteristic of exploiting all the potential offered by generative AI, through the use of two procedures: Fine-Tuning and Retrieval-Augmented Generation, which allow the system to offer good student support, such as the possibility of constructing concept maps and contextual summaries. We presented an initial experiment with a sample of 23 dyslexic students, coming from an Italian technical high school, obtaining positive feedback, along three lines: usability, engagement and perceived effectiveness. As future developments, we plan to carry out a more wide-ranging experiment to validate the system.

Acknowledgement. We would like to express our sincere gratitude to Universitas Mercatorum for its financial support in the development of this project (Grant No. 16-FIN/RIC 2023).

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